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(54) Title: AUTOMATED TENSIONING MACHINE FOR SPOKE WHEELS

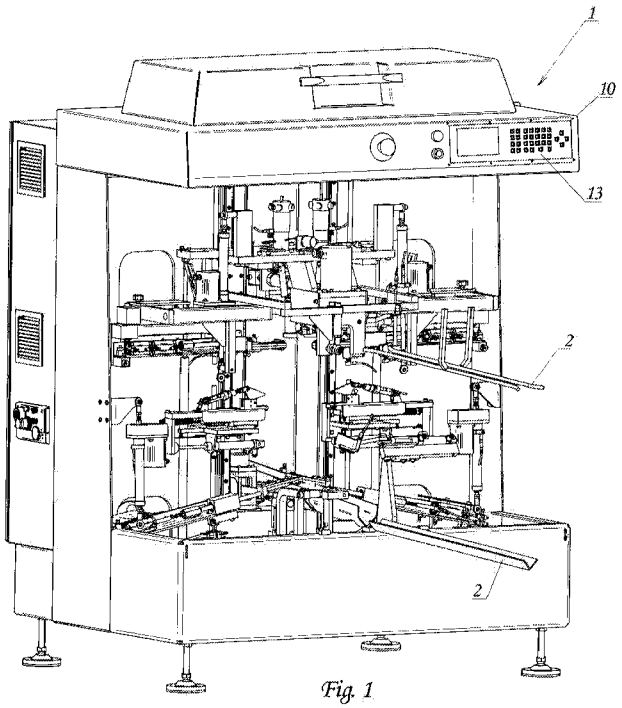


Fig. 1

(57) Abstract: There is disclosed a tensioning machine comprising means by which a spoke wheel provided with spokes to be tensioned between the wheel hub and the rim thereof, which spokes are provided with rotatable rim moving elements, is placed in the tensioning machine, fixing means by which the position of the wheel hub is fixed after the spoke wheel has been placed in the tensioning machine, driving means connected to the spoke wheel for rotating the spoke wheel, a control system for the tensioning machine, drivable spoke tensioning means connected to the control system, which spoke tensioning means can be connected to the rim moving elements, and measuring sensors connected to the control system, which measuring sensors are arranged for measuring any lateral and/or vertical deviation of the spoke wheel at least at the location of the rim. The control system is arranged for correcting the spoke tension in the spoke wheel on the basis of controlled - non-torque dependent - rotation of the rim moving elements and/or the spoke tensioning means.

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AUTOMATED TENSIONING MACHINE FOR SPOKE WHEELS

The present invention relates to a tensioning machine which is used for tensioning the spokes of a spoke wheel.

The present invention further relates to a method for the corresponding tensioning of spokes in spoke wheels.

In the tightening or tensioning machines that are known so far, tensioning of a separate laced - at low tension, spoke wheel whose spokes are to be tensioned yet, takes place. Using specially formed heavy clamping constructions comprising shell members that bed down on the rim, the rim and the wheel hub are firmly and immovably clamped down. The spokes are tensioned by tightening nipples provided on the spokes on the inner side or the outer side of the rim, whilst the position of the rim remains unchanged on account of the clamped position of the rim. The nipples form rotatable rim moving elements, which are tightened with a predetermined torque in the known machines. The idea is that said predetermined torque leads to a corresponding individual spoke tension and that the virtual absence of any lateral and/or vertical alignment deviation in the rim will be retained also after the clamping shell members are removed.

A drawback of those machines is the fact that different, dedicated shell members, which must be manufactured with a high degree of precision, are needed for every differently formed type of spoke wheel exhibiting different rim widths, spoke positions, rim shapes and diameters so as to be able to clamp down those rim types as well.

The object of the invention is to provide an improved, simpler and less heavy tensioning machine and a

corresponding method which are more universal and which require fewer machine adaptations in use and which furthermore provide more precisely aligned spoke wheels which, in the long run, will exhibit less lateral and/or vertical alignment deviations.

In order to accomplish that object, the tensioning machine and the tensioning method according to the invention exhibit the features summed up in independent claims 1 and 8.

The advantages of the tensioning machine and the tensioning method according to the invention include the fact that the rim of the spoke wheel to be tensioned no longer needs to be clamped down. Thus, the shell members required for doing so are no longer needed and it is no longer necessary to produce dedicated shell members for the rim type in question. Furthermore, converting the machine and exchanging shell members for shell members suitable for other types of spoke wheels, which was previously required, is no longer necessary, either, so that the machine can be more quickly adapted to the multitude of available rim sizes, spoke patterns and rim types.

Another embodiment of the tensioning machine is characterised in that the control system is arranged for measuring any lateral and/or vertical alignment deviations at a number of rim positions angularly symmetrically disposed around the wheel hub.

This leads to a gain in time in combination with the quick realisation of a desired uniform mechanical tension distribution at the desired tension level.

Further detailed possible embodiments as defined in the other dependent claims and their concomitant advantages will be set forth in the description below.

The tensioning machine and the related method according to the present invention will now be explained in more detail with reference to the following figures, in which like parts are indicated by the same numerals. In the drawing:

Figure 1 is a front view of a tensioning machine according to the invention;

Figure 2 is a detail view of the tensioning machine of figure 1; and

Figure 3 shows a part of a spoke wheel comprising spokes to be tensioned in the tensioning machine shown in figures 1 and 2.

Figures 1 and 2 schematically show the practical structure of a tensioning machine 1 and a detail thereof. The tensioning machine 1 comprises means 2 via which a spoke wheel 3 (partially shown in figure 3) can be placed in the tensioning machine and be discharged therefrom after the spokes 4, which are usually provided in non-tensioned condition in the spoke wheel 3, have been tensioned. The spokes 4, which extend between the wheel hub 5 and the rim 6, are provided with rotatable rim moving elements 7 on the side of the rim 6. In practice said elements are configured as nipples 7 mounted on a spoke end, which can be rotated from the inner side and/or from the outer side of the rim 6 so as to tension the spoke 4 to a desired mechanical tension by moving the rim 6.

The tensioning machine 1 further comprises fixing means 8, by means of which the wheel hub 5 can be fixed in position in the machine 1 after the spoke wheel 3 has been placed in the tensioning machine 1 via the means 2. The tensioning machine further comprises driving means 9 connected to the spoke wheel 3, usually configured as rotatable drive rollers to be pressed against the wheel 3, by means of which the spoke wheel 3 can be rotated as

desired after being fixed in position.

The tensioning machine 1 further comprises a microcomputer software controlled control system 10, for influencing the various means and elements already explained and yet to be explained, and in particular the order in which the various operations in the tensioning machine 1 are carried out. Connected to the control system 10 are, among other things: drivable spoke tensioning means, generally indicated by reference numeral 11, for example comprising rotatable nipple wrenches 11-1, which engage around the nipples 7 and spokes 4, or screwdrivers 11-2 which fit the nipples 7 on the ends of the spokes 4 - which nipples can generally be connected to the rim moving elements 7 - as well as measuring sensors 12-1, 12-2, for example optic sensors, which are arranged to measure any lateral and/or vertical alignment deviations of the spoke wheel 3, at least at the location of the rim. Furthermore, sensors 12-3 are provided for determining the position of the spokes 4 relative to the tensioning means 11.

The control system 10 is designed to correct the spoke tension in the spoke wheel by controlled rotation of the spoke tensioning means 11 independently of the torque being applied. In other words, the extent to which said means 11 are rotated is the measure for moving the rim 6 and tensioning the spokes 4 of the spoke wheel 3. This provides a direct relation between the extent of rotation and the associated movement of the means 11, and thus of the rim 6, of course after the nipple wrenches 11-1 and/or the screwdrivers 11-2 have first made a free turn. The consequence is that when all the spokes 4 have been tensioned to the correct tension by the spoke tensioning means 11, any lateral and/or vertical alignment deviations of the spoke wheel 3 will have been minimized.

The control system 10 is preferably designed to measure the lateral and/or vertical alignment deviation at a number of rim positions, which may be angularly

symmetrically disposed around the wheel hub 5, although this is not necessary, at which positions the spokes 4 are tensioned to the desired tension by tightening the nipples 7. Said rim positions, at which the spoke tensioning means 11 are disposed in that case, are arranged diametrically opposite each other, for example at an angle of 180 degrees, as shown in figures 1 and 2, at an angle of 120 degrees relative to each other or in a quadrant, viz. at an angle of 90 degrees relative to each other. According to a particular strategy, the rim moving elements 7 are rotated in a desired pattern, driven by the driving means 9, until a spoke wheel 3 is obtained whose spokes 4 all have the desired tension, with the wheel 3 exhibiting a minimal alignment deviation. A desired strategy for measuring and/or tensioning the spokes 4 in a desired order, which strategy is optimal with a view to achieving the desired final result, may be followed with every type of spoke wheel 3 or rim 6, making use of the aforesaid correction realised by rotating the elements or nipples 7.

According to another possible strategy, the control system 10 is for example programmed to measure the lateral and/or vertical alignment deviation at one or more rim positions and carry out the measurements anew after corrections have been made, and to use the latest measuring results for further corrections of the lateral and/or vertical alignment deviation, possibly at one or more other rim positions. Usually, the tensioning machine 1 is designed for determining the movement of the rim 6 of the spoke wheel 3 on the basis of a number of - possibly partial - revolutions of the spoke tensioning means 11 or the rim moving elements 7 during tensioning and/or on the basis of the movement of the spoke moving means 7 relative to the spoke 4. In the latter case, it is in fact the effective length of the spoke 4 between the wheel hub 5 and the inner side of the rim 6 that is measured.

In a further embodiment, the tensioning machine 1

comprises registration means 13 for recording the lateral alignment deviation and/or the vertical alignment deviation for each rim position, such as the rim position where the spokes engage, in a memory, usually a memory of the control system 10. In particular the changes in the various alignment deviations in the wheel 3 during tensioning can be constantly measured and stored, possibly following a predetermined strategy.

The aforesaid controlled rotation of the means 7 and/or the spoke tensioning means 7 takes place in dependence on the - possibly automatic - recognition of the type of spoke wheel 3, which can for example be carried out by means of a number of probes or sensors (not shown) which are known per se. Adaptation to the type of wheel 3 and positioning of the aforesaid means 7 and 11, which may be carried out in an automated or non-automated process, takes place along guides 14.

The controlled rotation of the means 7 and/or the spoke tensioning means 11 for realising the desired spoke tension may take place in dependence on the results of current measurements of the lateral and/or vertical alignment deviation.

CLAIMS

1. A tensioning machine comprising:
- means by which a spoke wheel provided with spokes to be tensioned between the wheel hub and the rim thereof, which spokes are provided with rotatable rim moving elements, is placed in the tensioning machine;
- fixing means by which the position of the wheel hub is fixed after the spoke wheel has been placed in the tensioning machine;
- driving means connected to the spoke wheel for rotating the spoke wheel;
- a control system for the tensioning machine;
- drivable spoke tensioning means connected to the control system, which spoke tensioning means can be connected to the rim moving elements;
- measuring sensors connected to the control system, which measuring sensors are arranged for measuring any lateral and/or vertical deviation of the spoke wheel at least at the location of the rim;
wherein said control system is arranged for correcting the spoke tension in the spoke wheel on the basis of controlled - non-torque dependent - rotation of the rim moving elements and/or the spoke tensioning means.

2. A tensioning machine according to claim 1, characterised in that the control system is arranged for measuring any lateral and/or vertical alignment deviations at a number of rim positions angularly symmetrically disposed around the wheel hub for additionally correcting the spoke tension on the basis thereof.

3. A tensioning machine according to claim 1 or 2, characterised in that the spoke tensioning means are arranged diametrically opposite each other, viz. at an angle of 180 degrees, at an angle of 120 degrees relative

to each other or in a quadrant, viz. at an angle of 90 degrees relative to each other.

4. A tensioning machine according to any one of claims 1-3, characterised in that the control system is designed to measure the lateral and/or vertical alignment deviation at one or more rim positions and carry out the measurements anew after corrections have been made, using the latest measuring results for further corrections of the lateral and/or vertical alignment deviation, possibly at one or more other rim positions.

5. A tensioning machine according to any one of claims 1-4, characterised in that the tensioning machine comprises registration means for recording the lateral alignment deviation and/or the vertical alignment deviation for each rim position, such as the rim position where the spokes engage, in a memory.

6. A tensioning machine according to any one of claims 1-5, characterised in that the means for placing the spoke wheel in the tensioning machine are designed to place, and possibly also discharge, the spoke wheel in an upright orientation.

7. A tensioning machine according to any one of claims 1-6, characterised in that the tensioning machine is designed for determining the movement of the rim of the spoke wheel on the basis of a number of - possibly partial - revolutions of the spoke tensioning means and/or the rim moving elements during tensioning and/or on the basis of the movement of the spoke moving means relative to the spoke.

8. A method wherein spokes in a spoke wheel provided with a rim are tensioned by means of rim moving elements and/or

spoke tensioning means, in a manner in which movement of the freely movable rim of the spoke wheel is effected by controlled - non-torque dependent - rotation of the aforesaid elements and/or means.

9. A method according to claim 8, characterised in that the lateral and/or vertical alignment deviation at one or more rim positions is measured and that said measurements are carried out anew after corrections have been made, using the latest measuring results for further corrections of the lateral and/or vertical alignment deviation, possibly at one or more other rim positions.

10. A method according to claim 8 or 9, characterised in that the movement of the rim of the spoke wheel is determined on the basis of a number of - possibly partial - revolutions of the spoke tensioning means and/or the rim moving elements during tensioning and/or on the basis of the movement of the spoke moving means relative to the spoke.

11. A method according to any one of claims 8-10, characterised in that said controlled rotation takes place in dependence on the, possibly automatic, recognition of the type of spoke wheel.

12. A method according to any one of claims 8-11, characterised in that measurement of the lateral and/or vertical alignment deviation takes place, possibly continuously, while the spoke wheel is being tensioned.

13. A method according to any one of claims 8-12, characterised in that said controlled rotation takes place in dependence on the results of current measurements of the lateral and/or vertical alignment deviation.

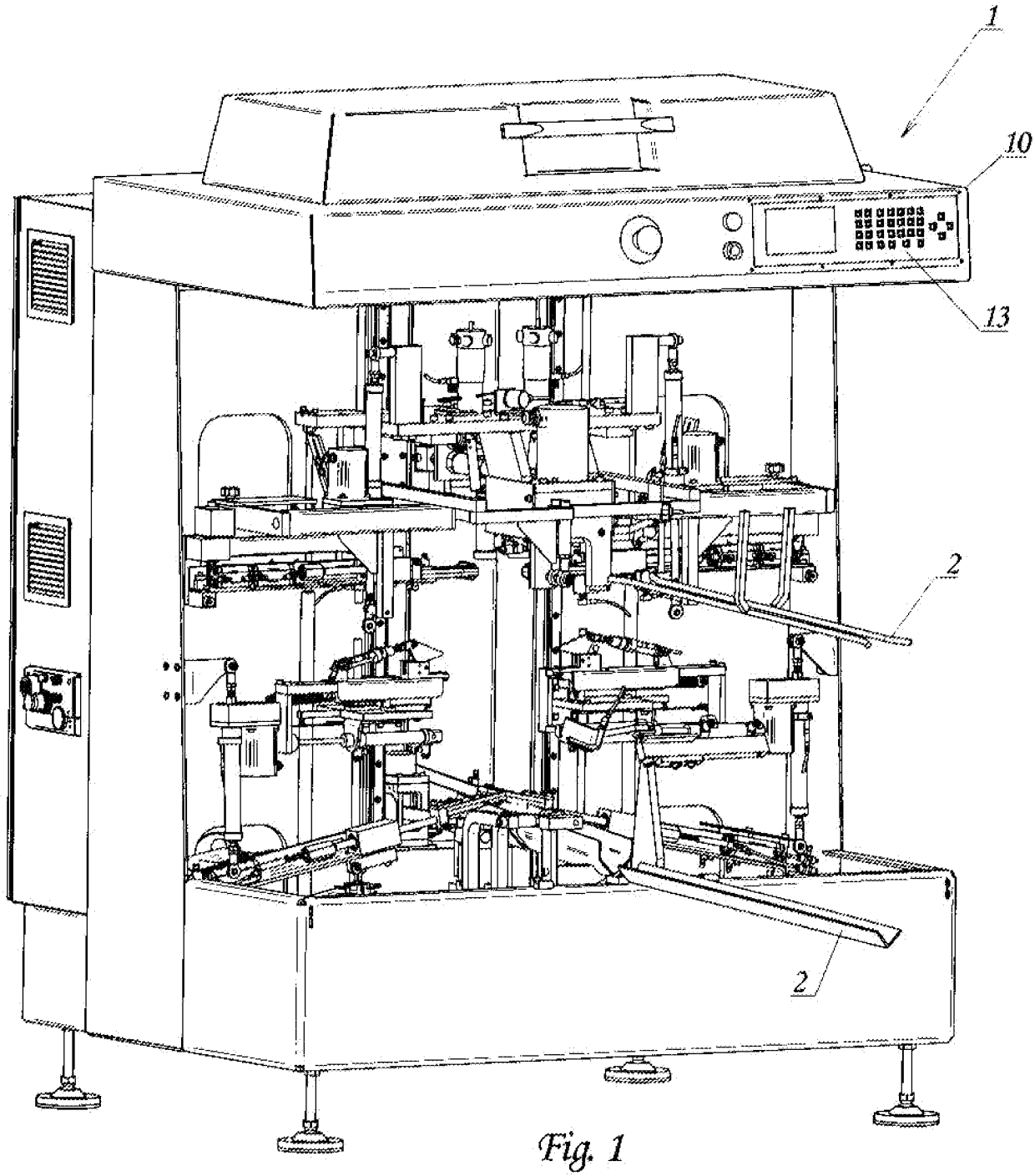


Fig. 1

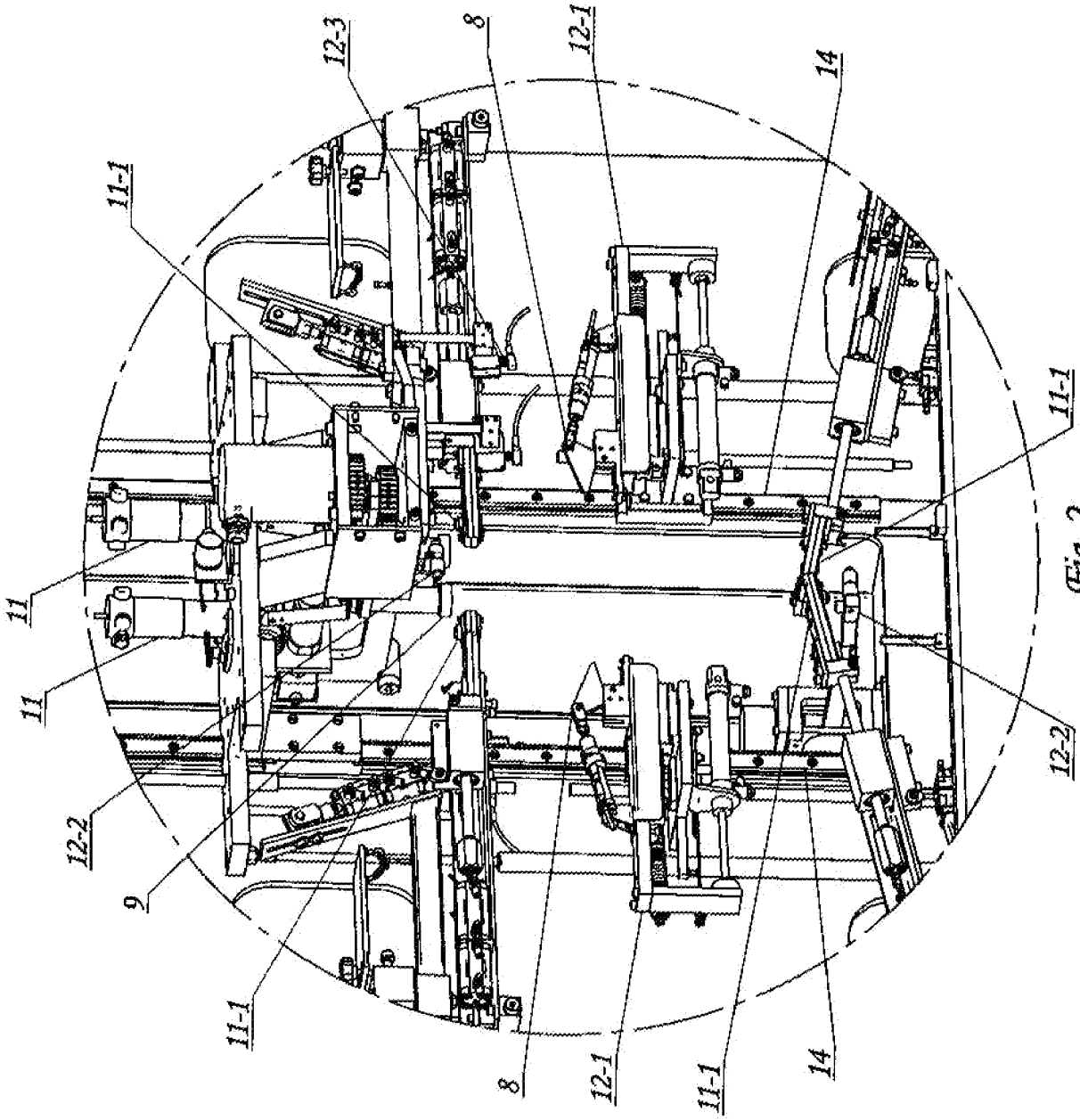


Fig. 2

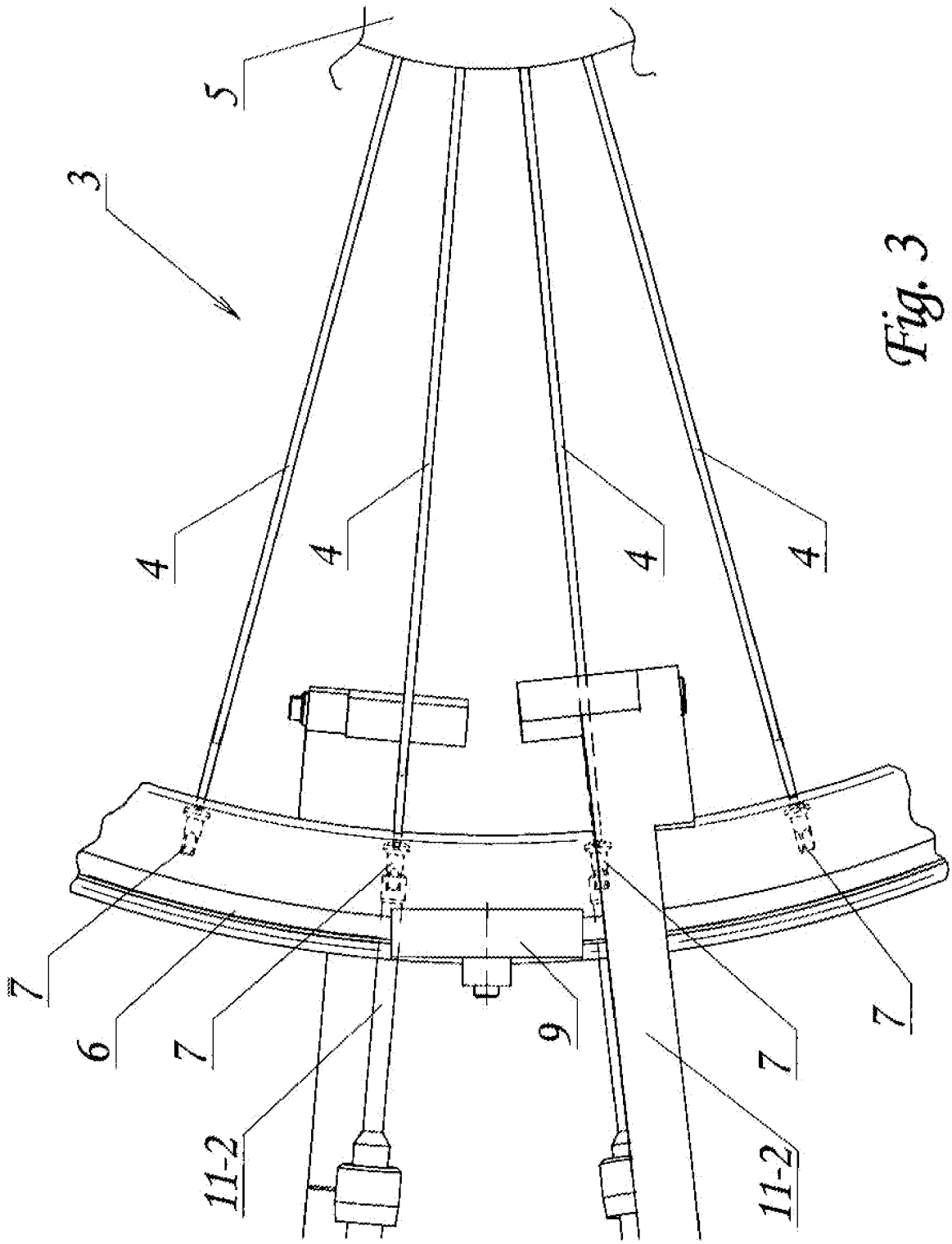


Fig. 3

INTERNATIONAL SEARCH REPORT

International application No
PCT/NL2010/050896

A. CLASSIFICATION OF SUBJECT MATTER
INV. B60B31/02
ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)
B60B

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practical, search terms used)
EPO-Internal, WPI Data

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	GB 856 691 A (TI GROUP SERVICES LTD) 21 December 1960 (1960-12-21) page 1, line 30 - page 2, line 15; figures 1-5	1-13
A	US 2 533 113 A (HENRY HARRY H) 5 December 1950 (1950-12-05) column 3, line 70 - column 4, line 70; figures 1,2	1,8
A	JP 61 006003 A (BRIDGESTONE CYCLE CO) 11 January 1986 (1986-01-11) * abstract	1,8
A	WO 00/16996 A1 (BEST MACHINE DEV V O F [NL]; BOSNJAK MLADDEN [NL]; BEST BOB [NL]) 30 March 2000 (2000-03-30) claims; figure 1	1,8

☐ Further documents are listed in the continuation of Box C.

☒ See patent family annex.

* Special categories of cited documents :

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"E" earlier document but published on or after the international filing date

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"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

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"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
10 March 2011	18/04/2011

Name and mailing address of the ISA/	Authorized officer
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INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No
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Patent document cited in search report	Publication date	Patent family member(s)	Publication date
GB 856691 A	21-12-1960	NONE	
US 2533113 A	05-12-1950	NONE	
JP 61006003 A	11-01-1986	JP 2562129 B2	11-12-1996
WO 0016996 A1	30-03-2000	EP 1121256 A1	08-08-2001
		NL 1010159 C1	24-03-2000